

Annotation Rules and Error Taxonomy

*The instrument for auditing plain-language summaries of
clinical trial results against their evidence tables.*

Taxonomy v0.4 — 17 codes, 6 of commission and 11 of omission

Annotation rules 4, with a correspondence rule and 6 resolved boundary cases

Derived from Regulation (EU) No 536/2014, Annex V; CTR Q&A; GLSP v1

Project lay-summary-audit · Karen Guimarães Hoyer

Extracted 19 September 2026

HOW TO READ THIS

Every rule here was written after reading an output that the taxonomy in force could not score. None was derived in the abstract, and each carries the source it was derived from and the consequence it has for the code.

The object under audit is a plain-language summary produced from a structured evidence table. The auditor receives both and judges the summary against the table; a human reviewer separately verifies the table against the article. Everything needed to score a code must be present in the evidence table, and a gap in the table is an extraction gap, not a generator failure.

THE DISTINCTION THE WHOLE INSTRUMENT RESTS ON

Fidelity and compliance are not the same axis. A summary can be accurate about its source and wrong about its obligations. It can reproduce every figure correctly and still omit a mandatory element, present a frequency in a form the reader cannot use, or treat a blocked outcome as demonstrated. Where a published source reports in a way the current norm disapproves, a faithful summary can inherit the non-compliance.

Codes of commission police fidelity and presentation. Codes of omission police completeness. A summary must pass both.

A TAXONOMY SCORES WHAT IT ENUMERATES

Five consecutive generator runs produced no finding while the taxonomy held six codes, all concerned with fidelity to the evidence table. The first summary scored against a taxonomy that also covered mandatory-element completeness failed on six counts. The generator had not changed; the ruler had. A clean result is evidence about the instrument as much as about the object.

PART I — THE TAXONOMY, V0.4

Codes of commission

Six codes. They concern what the summary asserts, and how it presents what it asserts. Each was derived from an observed case.

CODE	DEFINITION
C1	Unsupported efficacy claim. A claim of benefit the evidence table does not support. Arises from invention, and equally from misreading a result that is present — generalising from one outcome to a domain, for instance reporting improvement in a subjective complaint measure as improvement in cognition.
C2	Significance misstated. A result described as reliable, confirmed or demonstrated that the analysis plan does not sustain. Includes treating an outcome outside the confirmatory sequence as established, and treating an unadjusted secondary as confirmatory.
C3	Statistical apparatus printed as a number. A confidence interval, p value or test statistic given as a numeric value. Applies regardless of whether the value matches the evidence table.
C4	Quantitative error. A value, denominator, direction or arithmetic result that contradicts the evidence table. Applies in prose as well as in tables.
C5	Figure given for an outcome outside the confirmatory sequence. Any effect estimate, per-arm count or percentage for an outcome whose confirmatory testing was blocked. Qualifying language does not redeem it, and nominal favourability is not required for the code to apply.
C6	Frequency in a form that leaves a calculation to the reader. An adverse event frequency as a bare percentage, or a table of percentages without the denominator stated in words at the point of use. The value may be correct; the form is the error.

Codes of omission

Eleven codes. O1 concerns the qualification of uncertainty. **O2 to O11 map one-to-one onto the ten elements of Annex V**, and are scored against the generator only where the evidence table carries the information.

CODE	DEFINITION	ANNEX V
O1	Uncertainty not qualified. A non-significant or non-confirmatory result presented without conveying its unreliability. Not a general catch-all for omissions.	—
O2	Trial identification absent or incomplete — title, protocol number, EU trial number or other identifiers.	Element 1
O3	Sponsor absent — name or contact details.	Element 2
O4	General information absent or incomplete — where and when conducted, main objectives, reasons for conducting it.	Element 3
O5	Population absent or incomplete — numbers by region, age and gender breakdown, inclusion and exclusion criteria.	Element 4

CODE	DEFINITION	ANNEX V
06	Investigational medicinal products absent.	Element 5
07	Adverse reactions or their frequency absent. Where the source carries only a pre-specified subset, stating that limitation is compliant; failing to state it is not.	Element 6
08	Overall results absent.	Element 7
09	Comments on the outcome absent.	Element 8
010	Follow-up trials not indicated.	Element 9
011	Where additional information can be found absent.	Element 10

SCOPE OF AN OMISSION CODE

An element is **absent** when nothing in the summary addresses it, and **materially incomplete** when a sub-item named in the Annex V text is missing. Both are scored. An element the evidence table does not carry cannot be scored against the generator: the gap is in extraction, and is recorded as such.

DOI, registry URL and citation are good traceability practice and are not required elements. Their absence is not an omission under this taxonomy.

PART II — THE ANNOTATION RULES

Four rules. Each governs what a lay summary does, not what its source did.

RULE 1

Uncertainty is explained, not displayed

RULE

A confidence interval, p value or test statistic given as a numeric value is **C3**, regardless of whether it matches the evidence table. Uncertainty must be conveyed in words.

The quality check sits at the level of the quantitative statement, not at the level of the notation. A summary that prints a correct interval has not made an arithmetic error; it has handed the reader an object the reader cannot use, in place of the statement the reader needs.

Source · GLSP v1, numeracy principles · Expert Group guidance on summaries for laypersons, v2, February 2018.

RULE 2

An outcome outside the confirmatory sequence receives no effect estimate

RULE

Where the analysis plan blocked confirmatory testing of an outcome, the summary reports it as **measured**, with three qualifying elements, and gives no effect estimate. An effect estimate is **C5**. A missing qualifying element is **O1**.

The three qualifying elements

1. **Q1** — where the full results for the outcome can be found.
2. **Q2** — that it was not formally tested, and why.
3. **Q3** — that it does not support a conclusion about benefit.

Source · Regulation (EU) No 536/2014, Annex V · CTR Q&A, July 2026, chapter 6 (§6.1, §6.2), from which the qualifier "potentially" was removed in the secondary endpoint obligation.

RULE 3

No figures at all for a blocked outcome

RULE

Rule 2 forbids an effect estimate. Rule 3 extends it: a blocked outcome receives **no figures of any kind** – no effect estimate, no per-arm counts, no percentages. It is reported as measured, qualified per rule 2, and referred to the scientific summary.

The lay reader cannot interpret a raw count without the inferential frame the analysis plan withheld. Presenting one is presenting apparatus rather than information, which is the objection rule 1 makes to printing a confidence interval. What the reader needs is the conclusion.

Scope

This applies only to outcomes outside the confirmatory sequence. Counts for tested outcomes, for populations and for adverse events are required by Annex V and are unaffected.

EXTERNAL CONVERGENCE

NEJM Evidence editorial policy holds that in hierarchical testing procedures, p values should be reported only until the last comparison that was statistically significant, and that p values for the first non-significant comparison and all subsequent ones should not be reported. An independent source reaching the same position, applied to the scientific article rather than to the lay summary. Consulted 14 September 2026.

Derived after · reading a generated summary that withheld counts for a blocked renal composite. The rule was written to make that behaviour compliant by definition rather than by accident, and is declared as such.

RULE 4

Adverse event frequency as a natural frequency

RULE

An adverse event frequency is reported as a **natural frequency with an explicit denominator** – about 16 in every 100 people who took the intervention, against about 11 in every 100 who took the comparator. A bare percentage, or a table of percentages without the denominator in words, is **C6**.

Annex V requires a description of adverse reactions and their frequency, so omission is not an option and the question is form. GLSP numeracy principles require whole numbers, consistent denominators, and that no calculation be left to the reader. An isolated percentage leaves the calculation pending: how many people is that, and how large is the difference.

Distinction from rule 1

Confidence intervals and p values are inferential apparatus: without them the reader loses no fact. **An adverse event frequency is incidence, which is the fact itself.** Explain-rather-than-display applies to the first, not to the second.

Where denominators differ within a trial

The denominator in use is stated in words at the point of use, not only in a footnote.

Source · GLSP v1, numeracy principles · Annex V, element 6.

PART III — CORRESPONDENCE

What counts as a detection

THE CORRESPONDENCE RULE

A detection corresponds to the reference error when it carries the **correct code** and **localises the adulterated statement unambiguously**. Neither literal reproduction nor identical punctuation is required. For an omission, the auditor must identify the missing mandatory information and its reference evidence; it is not required to quote text that does not exist.

Three things are recorded separately for every detection: the code, the localisation or identification of the omission, and the combined correspondence.

The six boundary cases, resolved in advance

SITUATION	RESOLUTION
Partial quote	Accepted if it uniquely identifies the span. Uniqueness is verified per case and recorded in the manifest.
Correct localisation, wrong reference	Counts as a detection. The reference is marked incorrect.
Correct discrepancy, wrong code	Not a hit for the target code. Recorded separately as a classification error.
Additional flag	The hit is preserved. The extra flag is reviewed: if unfounded it is a false alert; if founded, the reference standard is revised and the revision is logged with a version.
Duplicate	Counted once.
O1 without a reference	Counted if the code and the element are unambiguous. Support is marked absent.

DENOMINATORS DO NOT MIX

A false alert raised on a positive case does **not** enter the specificity denominator. Any unfounded alert on a negative case makes that document a false positive. The two denominators are kept apart.

Revision of the reference standard

The original manifest is preserved and corrections are versioned. Where a founded additional flag reveals a second true error in a case, that case is excluded from the principal analysis and reported separately. No case is replaced on the basis of performance.

Multiple codes per finding

A single span can violate more than one code legitimately. A summary stating "improvement in cognition" where a subjective complaint measure improved and an objective processing-speed measure did not is **C1**, generalising from one outcome to a domain; if it also presents that as demonstrated where the secondary carried no multiplicity adjustment, it is **C2**. Both are correct about the same sentence.

DECISION

The manifest admits multiple codes per finding. A finding carries a list of codes, not one code.

Rejected alternatives. Precedence between codes is simpler and discards the information that the summary erred in two distinct ways at once. Redefining codes so they cannot overlap produces artificial categories that no longer correspond to the sources they were derived from.

Consequence for the correspondence rule

The boundary case "correct localisation, wrong code" assumed a single correct code, and is amended. Where the reference standard carries several codes for a finding, an auditor assigning a proper subset scores a detection hit and a partial classification hit. A code outside the reference set is an additional flag, handled by the rule above.

PART IV — SCORING

Two denominators, never pooled

Because the unit of the reference standard is the **finding**, and a finding may carry several codes, the denominators diverge and are reported separately.

DENOMINATOR	UNIT AND CRITERION	MEASURES
Per-finding	The number of findings. The auditor is correct on a finding if it localises it and assigns at least one of the reference codes.	Detection
Per-code	The number of (finding, code) pairs. The auditor is correct on a pair if it assigns that code at the correct location.	Classification

A single sensitivity figure over the panel is not well defined without stating which denominator it uses. **Detecting is not classifying.** An auditor that localises every problem and misassigns half the codes is useful; one that assigns codes well and misses localisations is not.

Agreement

Inter-rater kappa is withdrawn. A single annotator is declared, and kappa between raters has no two raters to measure. An optional intra-rater reassessment may be run on the same frozen outputs with shuffled identifiers, after two to three weeks, without access to the prior judgment; raw agreement and the disagreements are reported. Kappa is undefined where judgments are constant in one category.

What a reported figure may be read to mean

THE CONSTRAINT ON REPORTING

Sensitivity and specificity computed on this panel describe detection of **injected errors of known type in this construction**. Presenting them beside a caveat does not stop them being read as performance in a regulatory setting.

Where a number cannot bear the reading it will receive, the question is whether to report it at all, not how to caveat it. The same reasoning removed a computed figure from an internal consistency check that had produced an arithmetically correct and wholly inapplicable result: *a printed number is a number someone will cite*.

Independence

Variants built from one base text are not independent observations. A binomial interval over them assumes an independence that does not hold, and a narrow interval computed that way is more misleading than no interval. Where a code's cases derive from a single trial, the count of independent base texts — not the count of cases — bounds what can be claimed.

Exact bounds under perfect performance

Two-sided Clopper–Pearson lower bounds, assuming independence: $6/6 = 54.1\%$ $10/10 = 69.2\%$ $12/12 = 73.5\%$. The panel is exploratory and was not sized to estimate sensitivity with precision.

PART V — STANDING DECISIONS

Rulings taken once, recorded so they are not retaken differently.

C5 applies whether or not the figure is qualified

Any effect estimate for an outcome outside the confirmatory testing sequence is C5, even when accompanied by a caveat. Nominal favourability is not a condition: an outcome whose interval crosses the null and one whose interval excludes it are both blocked, and both attract the code.

O11 requires a general pointer

A referral to the scientific summary for one specific outcome does not discharge Annex V element 10. The element requires an indication of where additional information about the trial can be found, in general. A footer carrying the registration identifier and the primary publication satisfies it; an outcome-specific referral does not.

The evidence table records the source; it does not editorialise it

Where a source reports a figure that current norms disapprove of — nominal p values for comparisons after a hierarchy stopped, for instance — those values stay in the evidence table, labelled. The table is the reference standard, not a summary. Removing them would decide for the generator the very thing the case exists to test: the source offers the number, and the summary should not treat it as confirmation. With no number offered, there is no test.

Source practice may predate the norm

An article published before a guideline existed is not in breach of it. Much of the published corpus predates the convergence recorded under rule 3, and that is a finding about the corpus rather than about any one trial. A finding is scored against the taxonomy, not against the source's own practice: that a source did something does not make it compliant for a lay summary to repeat it.

A generator output is not a negative by origin

Every candidate negative is reviewed against the evidence table and the taxonomy before admission. The number of outputs rejected or corrected is recorded, and corrected versions are identified as such. Injected errors are not representative of the generator's spontaneous errors, and the two are never pooled.

THE TEN ELEMENTS OF ANNEX V

Verbatim from Regulation (EU) No 536/2014, Annex V, *Content of the summary of the results of the clinical trial for laypersons*. The omission codes O2 to O11 follow this order.

#	ELEMENT	SUB-ITEMS STATED IN THE TEXT
1	Clinical trial identification	title of the trial, protocol number, EU trial number, other identifiers
2	Name and contact details of the sponsor	—

#	ELEMENT	SUB-ITEMS STATED IN THE TEXT
3	General information about the clinical trial	where and when conducted; main objectives; explanation of the reasons for conducting it
4	Population of subjects	number of subjects in the Member State concerned, in the Union and in third countries; age group breakdown; gender breakdown; inclusion and exclusion criteria
5	Investigational medicinal products used	—
6	Description of adverse reactions and their frequency	—
7	Overall results of the clinical trial	—
8	Comments on the outcome of the clinical trial	—
9	Indication if follow-up clinical trials are foreseen	—
10	Indication where additional information could be found	—

NORMATIVE SOURCES, PINNED BY VERSION

SOURCE	VERSION AND USE
Regulation (EU) No 536/2014	Annex V. The ten mandatory elements, and the origin of omission codes O2 to O11.
CTR Questions & Answers	July 2026, chapter 6 (§6.1, §6.2). The qualifier "potentially" was removed from the secondary endpoint obligation in this version.
Summaries of Clinical Trial Results for Laypersons	Expert Group, version 2, February 2018.
Good Lay Summary Practice, v1	October 2021. Predates the current Q&A and cites it in draft form. Numeracy principles underpin rules 1 and 4.
NEJM Evidence editorial policy	Consulted 14 September 2026. Not normative for this project; recorded as external convergence with rule 3.

END OF EXTRACT

What this document does not contain

NO RESULT

Stages 1 and 2 of the pipeline are implemented and run on development cases. The case panel is designed and has not been executed. No metric has been computed and no result is reported.

RULES ONLY

The findings log, the case manifest, the coverage matrix and the evidence tables are held in the project repository and are not reproduced here. Where a rule names a case, the case is recorded there with its provenance.

PROVENANCE

Every pipeline output records the prompt content hash, the evidence table hash, the HEAD commit and the working tree state at the moment of execution. Outputs that enter a denominator are versioned; scratch output is not.

lay-summary-audit · evaluation protocol, reference extract

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